

BPv7 Custody Transfer and Compressed Status Reporting

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Agenda

- Motivation / Constraints / Status
- Bundle Sequence Identification – General Concepts
- Custody Transfer
- Compressed Bundle Reporting
- CBR-based End-to-End Services

Motivation / Constraints / Status

- Efficient Reporting on Bundle Status (eg, delivery) required in space mission scenarios
- Custody Transfer Mechanism allowing 'opportunistic' custody
- Focus on immediate needs (EO missions, LunaNet) – suitable for small, tightly-managed networks or single mission scenarios
- CCSDS Orange Book 90% ready – publication in 2025 intended
- A lot of choices are 'left to implementation' – still the specification should provide a basic level of interoperability (with providing sufficient flexibility for different use cases)
- Early versions of this specification have been prototyped and evaluated

Bundle Sequence Identification – General Concepts

Bundle Identification

- Identification of sequences of bundles by **bundle sequence numbers** qualified by a **bundle sequence ID**
- Bundle Sequence Numbers are used in extension blocks
- Unique Identification of bundles (during bundle lifetime) by
 - Bundle Sequence ID $\neq 0$:
Block Source Administrative Endpoint ID* (eid) + Bundle Sequence ID (unsigned int) + Bundle Sequence Number (unsigned int)
 - Bundle Sequence ID $= 0$:
Block Source Administrative Endpoint ID* (eid) + Destination Endpoint ID + Bundle Sequence Number (unsigned int)

* Block Source Administrative Endpoint ID may be omitted if it is the same as the source node ID in primary block.

Bundle Identification - Remarks

- Handling of bundle sequence number counters (by IDs / destination endpoint IDs) done within administrative element
- Assignment of bundles IDs is use case specific, eg
 - service number or bundle originating service if ipn naming is used
 - depending on next hop (eg, for reception reporting)
 - destination endpoint ID (sequence ID '0'), eg for delivery reporting

Bundle Sequences / Bundle Sequence Collections

A **Bundle Sequence** shall be a CBOR definite-length array of 3 or 4 items:

- **First Sequence Number** - CBOR unsigned integer; this indicates the first Bundle Sequence Number of the bundles the bundle sequence is referring to
- **Bundle Sequence Length** - CBOR unsigned integer; this indicates the number of bundles that the bundle sequence is referring to, i.e. the sequence refers to all bundles with sequence number 'First Sequence Number' to 'First Sequence Number + Sequence Length -1'
- **The Bundle Sequence ID** - CBOR unsigned integer OR the common **Destination Endpoint ID** of the bundles in the sequence otherwise - endpoint ID
- Optional: **Block Source Administrative Endpoint ID** - endpoint ID

A **Bundle Sequence Collection** shall be a CBOR definite-length array of Bundle Sequence(s).

Custody Transfer

Custody Transfer Extension Block (CTEB)

- Custody Transfer is requested through a **Custody Transfer Extension Block** with the following block-type-specific data
 - **Bundle Sequence Number** – CBOR unsigned integer
 - **Bundle Sequence ID** – CBOR unsigned integer
 - **Block Source Administrative Endpoint ID** - endpoint ID (“current custodian”)
- A bundle shall never contain more than one Custody Transfer Extension Block.

Compressed Custody Signal (CCS)

- A Compressed Custody Signal is an **administrative record** containing a CBOR map composed by key/value pairs as record content:
 - Keys of the map shall be the Disposition Codes of the Custody Signal:
 - **Custody Acceptance**: a positive CBOR integer:
 - 1 – Custody Accepted (no further information)
 - Values above 1 – reserved for future use
 - **Custody refusal**: a negative CBOR integer
 - -1 – custody refused (no further information)
 - Values below -1 – reserved for future use
 - Values shall be **Bundle Sequence Collections***

*Bundle Sequences in the Bundle Sequence Collections above shall never contain a Block Source Administrative Endpoint ID (as the signal will be received by that administrative element)

Custody Procedures – Custody Decision

- Any node can **request custody** transfer by inserting a CTEB into a bundle.
- Any node receiving a CTEB which is not the destination of a received bundle shall **decide whether to take or refuse custody** of the received bundle.
- The decision whether to take custody are mission or network-specific:
 - A node may take custody of a bundle when there are ‘reasonable good chances’ that the bundle can be forwarded to a next custodian. This can be based e.g. on a combination of bundle time-to-live, knowledge about upcoming contacts, past experiences, etc.
 - By local policy (which could be set by network management)
 - Other means
- A node receiving a CTEB which is delivering the received bundle shall always take custody of the received bundle.

Custody Procedures – Custody Processing

- In case the node **takes custody**, it shall:
 - store all required information for creating a compressed custody acceptance signal
 - **delete the current CTEB**
 - before forwarding the bundle, **add a new CTEB** requesting custody transfer with its own Administrative Endpoint ID as the Block Source Administrative Endpoint ID
 - store the bundle until a custody acceptance signal is received or the bundle life-time has expired.
- In case the node **refuses custody** it shall store all required information for creating a compressed custody refusal signal.

Custody Procedures – Custody Signalling

- A Compressed Custody Signal shall be sent based on the stored information according to the MIB configuration or based on external events.
- The decision when to transmit a Compressed Custody Signal is an implementation matter:
 - Implementation may transmit Compressed Custody Signals **periodically based on a timer** if there are any CTEB requests to respond to.
 - Implementations may transmit Compressed Custody Signals **depending on size constraints** like the overall report bundle size or the number of bundles to report on. In this case, it is recommended that an additional timer is used to transmit a Compressed Custody Signal even if the size limit is not reached but no additional bundles to report on have been received for some time.
 - Implementations may transmit Compressed Custody Signals based on **external events** (e.g., the establishment of a communication link or reception of an explicit command).

Custody Procedures – Bundle Re-forwarding

- In case no Compressed Custody Signal acceptance signal is received by a custodian for a specific bundle, the bundle should be re-forwarded.
- The decision when to re-forward a bundle is an implementation decision. It may be based on:
 - **Timers**: If no custody signal for a bundle or is received for a certain amount of time, the bundle will be re-forwarded
 - **Received Custody Signals**: if in-sequence reception of bundles and custody signals can be assumed, gaps in the reporting of accepted custody signals can indicate lost bundles and lead to re-forwarding
 - **Explicit command**: based on an explicit command, all or certain bundles with pending custody acceptance may be re-forwarded
 - **Other Internal/External events** such as the initiation of a contact
 - **Reported failure** to forward the bundle via a Convergence Layer Adapter

Compressed Bundle Reporting

Compressed Reporting Extension Block (CREB)

- Compressed reporting is requested through a **Compressed Reporting Extension Block** with a definite-length CBOR array of 1 to 5 elements as block-type-specific data
 - Length 1: sequence number only
 - Length 2: sequence number with sequence ID
 - Length 3: reporting request by bundle source
 - Length 4: reporting request
 - Length 5: reporting request with report-to-eid
- A bundle shall may contain more than one Compressed Reporting Extension Block.

Compressed Reporting Extension Block (CREB) – Full Array Content

- **Bundle Sequence Number** – CBOR unsigned integer
- **Bundle Sequence ID** – CBOR unsigned integer (defaults to '0')
- **Status Report Request Flags** – CBOR unsigned integer (for report requesting)
 - Bit 0 – reception reporting request
 - Bit 1 – forwarding reporting request
 - Bit 2 – delivery reporting request
 - Bit 3 – deletion reporting request
 - Bit 4 – fragmentation reporting request
 - Bit 5 – reassembly reporting request
 - Bit 6 – custody acceptance reporting request
 - Bit 7 – custody rejection reporting request
- **Block Source Administrative Endpoint ID** - endpoint ID (defaults to administrative endpoint ID of bundle source node)
- **Report-to Endpoint ID**

Compressed Reporting Signal (CRS)

- A Compressed Reporting Signal is an administrative record containing a CBOR map composed by key/value pairs as record content:
 - Keys of the map shall be the Status Report Reason - CBOR unsigned integer providing the reporting reason.
 - Bit 0 – reception reporting request
 - Bit 1 – forwarding reporting request
 - Bit 2 – delivery reporting request
 - Bit 3 – deletion reporting request
 - Bit 4 – fragmentation reporting request
 - Bit 5 - reassembly reporting request
 - Bit 6 – custody acceptance reporting request
 - Bit 7 – custody rejection reporting request
 - Values shall be **Bundle Sequence Collections**

Compressed Reporting Processing

- Reception of a CREB with a **three, four or five-element array** shall cause the receiving bundle node to store all required information for generating a Compressed Reporting Signals.
- Compressed Reporting Signals shall be generated and transmitted to a common Compressed Reporting Signal Destination based on the stored information.
 - The decision when to transmit a Compressed Reporting Signal is an implementation matter.
 - Implementations may transmit Compressed Reporting Signals **periodically** if there is any information to report.
 - Implementations may transmit Compressed Reporting Signals depending on **size constraints** such as the overall report bundle size or the number of bundles to report on.
- Compressed Reporting Signal Destination shall be:
 - the **Source Node ID in the primary block** (three-element array)
 - the **Block Source Administrative Endpoint ID** (four-element array)
 - the **Report-To-Endpoint** (five-element array)

Bundle Fragmentation

- Custody Transfer Extension Blocks / Compressed Reporting Extension Blocks shall only be added to **bundles which have the 'Bundle must not be fragmented' Bundle Processing Control Flag set to true.**

CBR-based End-to-End Services

E2E Services – without CRS

- The following services can be realised with CREB containing just a sequence number per destination endpoint Id (sequence Id '0')
 - **E2E in-sequence Bundle Delivery**
 - Re-ordering at destination node according to sequence number
 - **Duplicate suppression**
 - Deletion of bundles with the same sequence number
 - **Gap Detection**
 - Identify gaps (or late bundles) by missing sequence numbers

E2E Services – with CRS

- The following services can be realised with the use of Compressed Reporting
 - **E2E Accounting**
 - Delivery Reporting by destination node (must support compressed reporting)
 - Deletion Reporting by deleting node (must support compressed reporting)
 - Reporting to source node a different report-to node
 - **E2E Reliability**
 - Based on E2E Accounting
 - Resending of bundles in case deletion reports are received or expected delivery reports are not received